MUHILARASAN T

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in Data Analytics, Data Science, and Machine Learning using Python and SQL. Skilled in data preprocessing, exploratory data analysis, visualization, statistical analysis, machine learning, model evaluation, and deployment. Experienced with Power BI, Pandas, NumPy, Scikit-learn, XGBoost, FastAPI, and SHAP through academic and portfolio projects.

TECHNICAL SKILLS

Languages: Python, SQL, C++, C
Data Analytics: Pandas, NumPy, Data Preprocessing, Exploratory Data Analysis, Data Cleaning, Statistical Analysis
Visualization: Power BI, Tableau, Matplotlib
Machine Learning: Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Model Evaluation
Models: XGBoost, Random Forest, HistGradientBoosting, KMeans
Deployment & MLOps: FastAPI, REST APIs, Docker, ML Pipelines, Model Deployment, Model Versioning, Basic CI/CD
Database & Tools: MySQL, Git, GitHub, Jupyter Notebook

PROJECTS

ARAN — Real-Time API Bot Detection & Automated Mitigation

Python | Pandas | Scikit-learn | XGBoost | FastAPI | SHAP

• Built an API bot detection system using 5,000 sessions and 13 behavioral features to distinguish legitimate users from automated traffic.
• Performed data preprocessing and trained an XGBoost classifier using cross-validation, hyperparameter tuning, and leakage-aware preprocessing.
• Implemented FastAPI real-time inference with SHAP-based explanations and automated API testing.

HABITA — Personalized Urban Housing Recommendation & Location Intelligence

Python | Pandas | Scikit-learn | HistGradientBoosting | KMeans

• Developed a housing recommendation system covering 55 Bangalore localities, combining rent prediction, commute prediction, and locality clustering.
• Built regression models for rent and commute estimation, achieving R^2 scores of 0.94 and 0.84 respectively.
• Applied KMeans clustering to group localities based on housing and lifestyle characteristics and integrated the results into personalized recommendations.

RED FLAG — Transaction Anomaly & Fraud Detection

Python | Pandas | Scikit-learn | XGBoost | FastAPI | SHAP

• Developed an end-to-end fraud detection platform for 1.85M transactions using a strict chronological train-validation-test strategy to prevent temporal leakage.
• Engineered 26 point-in-time-safe features and trained an XGBoost classifier, achieving 0.91 PR-AUC on an unseen test set.
• Optimized the classification threshold to reduce false-positive alerts and deployed FastAPI real-time and batch scoring with SHAP-based explainability.

CredX — Loan Risk & Repayment Analytics Platform

Python | MySQL 8.0 Star Schema | Power BI | DAX | SQL

• Modeled 39,717 loan contracts ($434.8M capital) into a normalized Kimball Star Schema on MySQL 8.0.
• Developed 8 advanced SQL analytical modules evaluating default hazard curves, credit grade risk stratification, and recovery rates.
• Constructed an interactive Power BI dashboard with 25+ DAX measures visualizing cash flow yields and delinquency drivers.

EXPERIENCE

Data Analytics Intern — Techvolt Software Pvt. Ltd., Coimbatore

• Performed data preprocessing, analysis, and visualization using Power BI.
• Created dashboards and reports to communicate data-driven insights and support decision-making.

EDUCATION

B.E. Computer Science and Engineering

Dhanalakshmi Srinivasan Engineering College, Perambalur | 2022 – 2026 | CGPA: 8.6

ACHIEVEMENT

First Place — Data Analytics Hackathon, NIT Trichy | March 2025

CERTIFICATIONS

• Masters Program in Data Science — NASSCOM (Gold, 96%), Sep 2026
• Microsoft Power BI Data Analyst Associate
• SQL Basic Course — Newton School
• SQL for Data Science — Coursera
• Data Analytics Job Simulation — Deloitte
• Data Science & Analytics — HP LIFE
• Python for Data Science — Udemy

LANGUAGES

Tamil | English